

AI Agents & Agentic AI Fundamentals

Objective

Understand:

- What is an AI Agent?
- Agent vs RAG
- Agentic AI Architecture
- Planning, Reasoning & Tool Calling
- Multi-Agent Systems
- Azure AI Foundry Agents
- Copilot Studio Agents
- Enterprise Agent Patterns

This is the bridge from:

Prompt Engineering

→ RAG

→ AI Agents

→ Enterprise AI Solution Architect

Learning Outcomes

By end of Day 5 you should be able to:

- ✓ Explain what an AI Agent is
- ✓ Explain Agent vs Copilot vs RAG
- ✓ Design Agent Architecture
- ✓ Identify enterprise use cases

- ✓ Understand Tool Calling
 - ✓ Understand Multi-Agent Systems
 - ✓ Draw Agentic AI Architecture
 - ✓ Prepare for Azure AI Foundry Agents and Microsoft Agent Framework learning
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Topic 1 – What is an AI Agent?

Traditional AI:

User

↓

GPT

↓

Answer

AI Agent:

User

↓

Agent

↓

Reason

↓

Use Tool

↓

Fetch Data

↓

Execute Action

↓

Answer

An Agent can:

- Think
- Plan
- Remember
- Use Tools
- Perform Actions

Unlike a normal chatbot.

Topic 2 – Agent vs RAG

Feature	RAG	Agent
Retrieve Data	✓	✓
Memory	✗	✓
Planning	✗	✓
Tool Calling	✗	✓
Multi-Step Execution	✗	✓
Take Actions	✗	✓

RAG Example

Question:

What is leave policy?

Search documents

Return answer

Agent Example

User:

Apply leave tomorrow

Agent:

Check balance

Create request

Notify manager

Send confirmation

Topic 3 – Core Components of an Agent

1. LLM

Brain

Examples:

- GPT-4o
- GPT-4.1

2. Memory

Stores:

- Context
 - Previous conversations
 - User preferences
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3. Tools

Examples:

SQL Database

Email

CRM

SharePoint

Azure APIs

Teams

GitHub

4. Planner

Determines:

Step 1

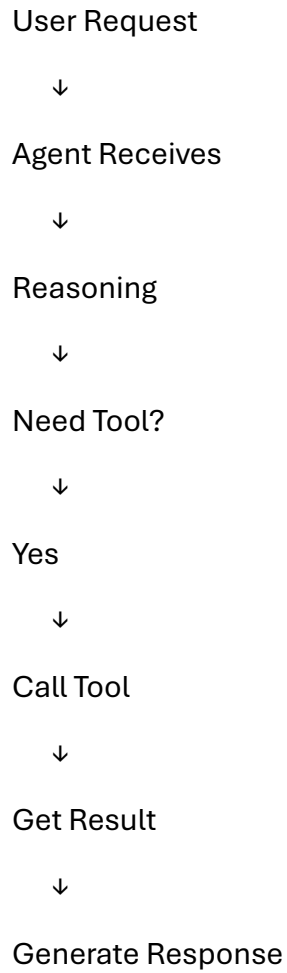
Step 2

Step 3

5. Executor

Runs actions.

Topic 4 – Agent Workflow



Topic 5 – Tool Calling

This is what makes Agents powerful.

Example:

User:

Show my open incidents

Agent:

Call ServiceNow API

Retrieve Incidents

Format Response

Another example:

User:

Create DevOps task

Agent:

Azure DevOps API

Create Work Item

Return ID

Topic 6 – Enterprise Agent Use Cases

HR Agent

- Leave
- Policies
- Benefits

IT Support Agent

- Incidents
- Password Reset
- Asset Requests

ESG Agent

Relevant to your current work:

Documents

Runbooks

ESG Reports

Master Data

Agent can answer and execute workflows.

Topic 7 – Multi-Agent Architecture

Instead of one super agent:

Planner Agent

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Research Agent

↓

Data Agent

↓

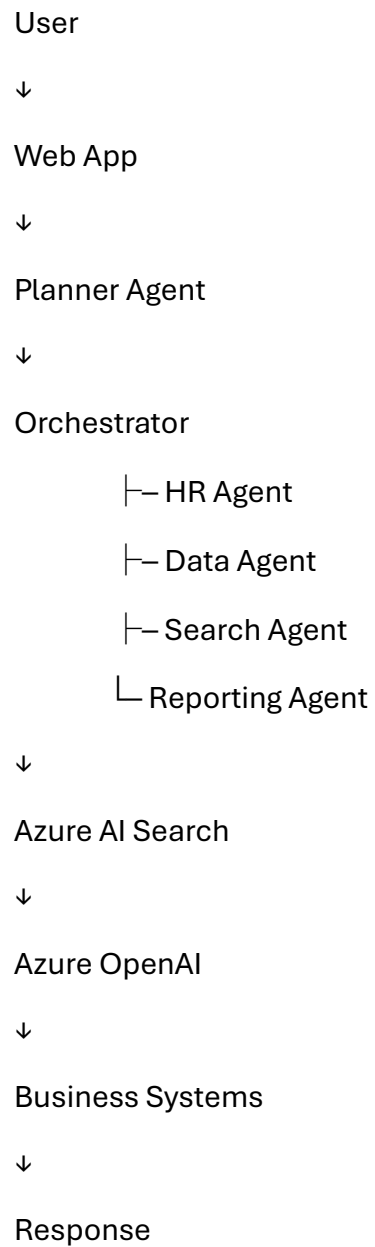
Reporting Agent

Benefits:

- ✓ Better accuracy
- ✓ Easier maintenance
- ✓ Scalability
- ✓ Domain specialization

Enterprise architectures are increasingly moving toward multi-agent orchestration.

Topic 8 – Agentic AI Architecture



Architect Focus:

- Security
 - Monitoring
 - Cost
 - Governance
 - RBAC
 - Responsible AI
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Topic 9 – Azure AI Foundry Agents

Learn Conceptually:

Models

Agents

Prompt Flow

Evaluations

Safety

Monitoring

Azure AI Foundry includes agent capabilities and monitoring capabilities for enterprise AI workloads.

Topic 10 – Copilot Studio Agents

Low-Code Agent Platform

Use Cases:

- HR Copilot
- IT Helpdesk Agent
- Knowledge Assistant
- Employee Onboarding Agent

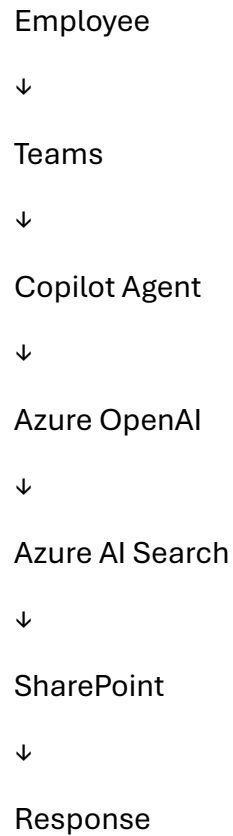
As an architect:

Focus on:

- Integrations
- Security
- Governance
- Data Sources

Architecture Assignment

Create Draw.io Diagram



Add:

- RBAC
- Managed Identity
- Monitoring

Hands-On Lab (Architect Track)

Build an Employee Exit Assistant Design

Requirements:

- Retrieve exit policy
- Retrieve handover checklist
- Retrieve asset return checklist
- Generate email draft
- Create ticket request

Design Architecture Only.

No coding required.